

Feature

CWRU Vibration
12 kHz DE · 48 recordings

Sliding Window
1024 pt · stride 512

28-D Features⁽¹⁾
10 time + 10 freq + 8 DE

Engineering

Training

Teacher 1D-CNN
48 K params · SA · FC

VRM-KD Distillation
 $\tau=4.0, \alpha=0.3, \lambda=0.5$

(2)–(3)

Compilation

Student KAN
[28,16,4] · 6148 params

NeuroPLC Compiler
IR → Optimizer → SCL

(4)

Deployment

S7-1200 CPU 1211C
40.3 KB / 75 KB · 7 388 FLOPs · ~22 ms

(5)